Example 3.12

Simplify $(5x+3)(5x+4)$ by using $(x+a)\ (x+b)$ identity.

Solution:

We have $(5x+3)(5x+4)$

Comparing it with $(x+a)(x+b)$, we get $x=5x$ and $a=3, b=4$

We know $(x+a)(x+b)=x^2+(a+b)x+ab$ (replacing x, a and b values)

$$(5x+3)(5x+4) = (5x)^2 + (3+4)(5x) + (3)(4)$$
$$= 5^2x^2 + (7)(5x) + 12$$
$$(5x+3)(5x+4) = 25x^2 + 35x + 12$$

Try these

Expand using appropriate identities.

(i) $(3p+2q)^2$ (ii) $(105)^2$ (iii) $(2x-5d)^2$

(iv) $(98)^2$ (v) $(y-5)(y+5)$ (vi) $(3x)^2-5^2$

(vii) $(2m+n)(2m+p)$ (viii) 203×197

(ix) Find the area of the square whose side is $(x-2)$ units.

(x) Find the area of the rectangle whose length and breadth are $(y+4)$ units and $(y-3)$ units.

3.6 Cubic Identities

I. $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

We shall prove it now,

$$LHS = (a+b)^3$$
$$= \big[(a+b)(a+b)\big](a+b) \text{ (expanded form)}$$
$$= (a+b)^2(a+b)$$
$$= (a^2+2ab+b^2)(a+b) \text{ (using identity)}$$
$$= a(a^2+2ab+b^2) + b(a^2+2ab+b^2) \text{ (using distributive law)}$$
$$= a^3 + 2a^2b + ab^2 + ba^2 + 2ab^2 + b^3$$
$$= a^3 + (2a^2b + ba^2) + (ab^2 + 2ab^2) + b^3 \text{ (grouping 'like' terms)}$$
$$= a^3 + 3a^2b + 3ab^2 + b^3$$
$$= RHS$$

$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

Hence, we proved the cubic identity by direct multiplication.

Aliter

$\times$	a^2	$2ab$	b^2
a	a^3	$2a^2b$	ab^2
b	a^2b	$2ab^2$	b^3

$= a^3 + 3a^2b + 3ab^2 + b^3$

Activity

You can visualize the geometrical proof of $(a+b)^3$ with the help of your teacher.

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